

Forecasting Credit Card Spending for Personalized Customer Relationship Management

A Predictive Analysis

Jessenth Ebenezer (15154)

The problem

Context

Accurate predictions of consumer spending patterns are vital for banks, enabling them to tailor their services and marketing strategies effectively.

Data

To achieve this, we will explore customer demographics, financial information, and past transaction history to build a predictive model.

Problem statement

Build a suitable Machine Learning model that predicts the average credit card spend of a customer over the next 3 months based on parameters listed in the dataset.

Challenges deep-dive

Data Quality

Missing Values and Irrelevant Features

As with any data of a significant size, there is a lot of information that may prove to be unwanted for the task at hand, implementing the right strategies to impute these values and taking decisions based on domain expertise is vital

Feature Complexity

Selecting and Engineering the right parameters

The complexity of financial data requires careful feature engineering to capture meaningful information. Deciding how to approach certain values require studying correlation and pattern recognition to ensure nothing of importance is lost and maintain accuracy throughout.

Model Selection

Going beyond thresholds

Going through a plethora of regression models and evaluating each model's performance and tuning them to make sure the expected performance criteria is met successfully, with various methods of ensemble learning, hyperparameter tuning, etc.

Solution

A robust model that follows the ensemble approach to provide reliable results

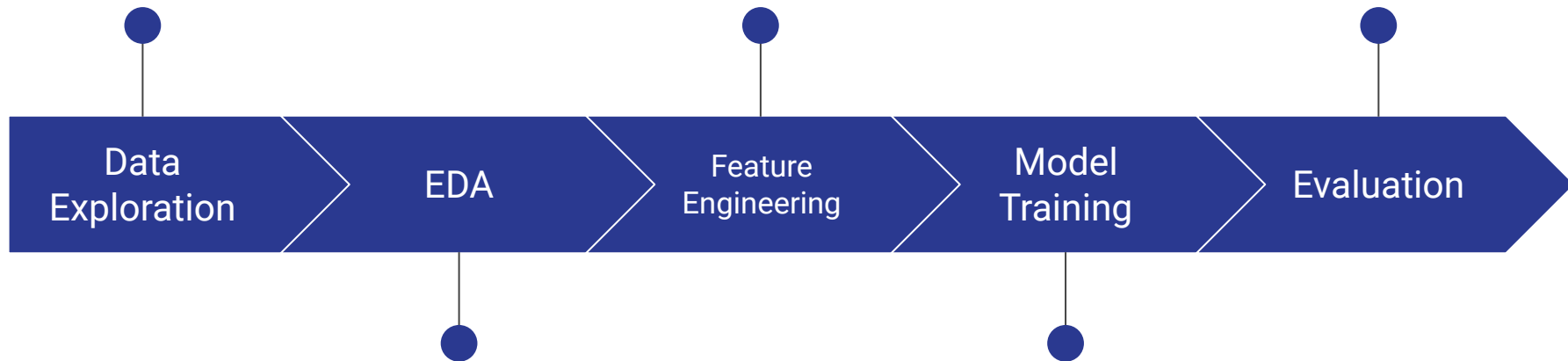
- Combining predictions from multiple base models such as Ridge, Lasso, Random Forest, Linear Regression, XGBoost, etc.
 - Scores well below threshold (~111 on validation set, 116-118 on evaluation dataset)
 - Can be easily deployed as a web application using gradio/streamlit
-

Implementation

Understanding the features and descriptions from provided dictionary, dataset and planning the forthcoming stages of analysis and prediction

Selecting features, combining and aggregating less correlated features, simplifying and cleaning data

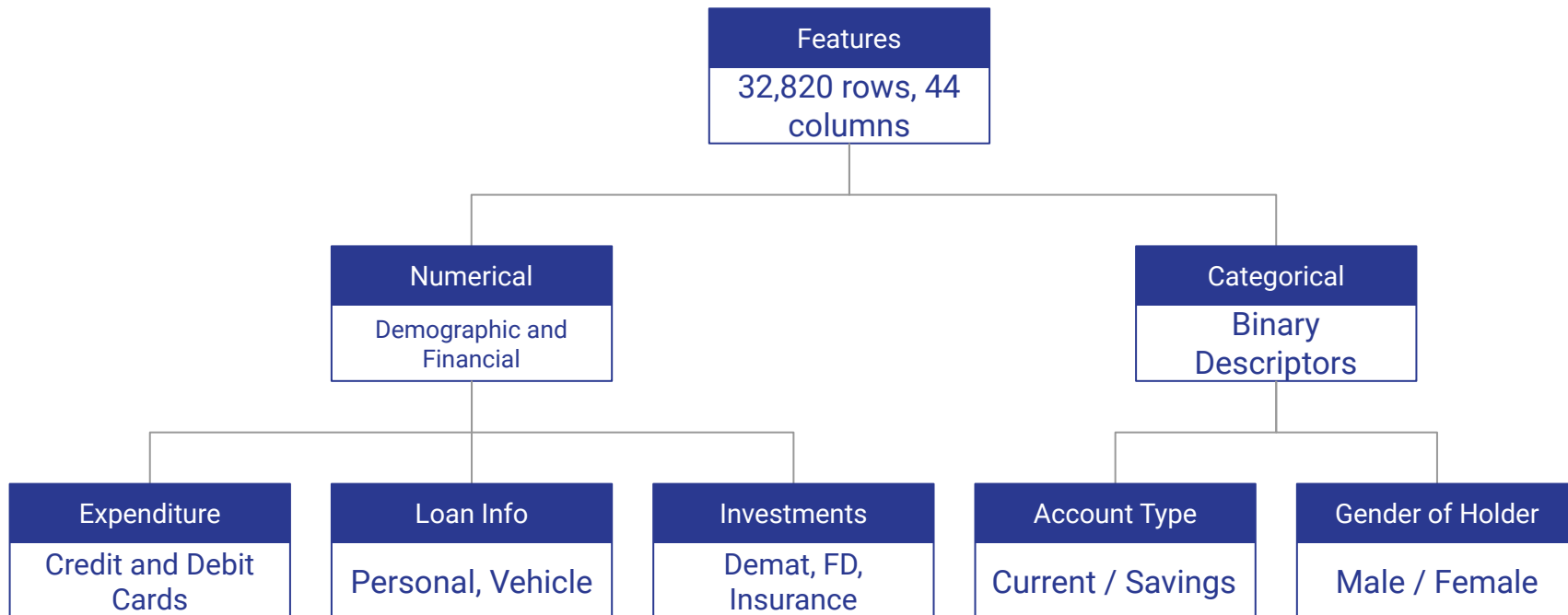
Comparing model efficiency and stacking to produce the ultimate meta model for prediction



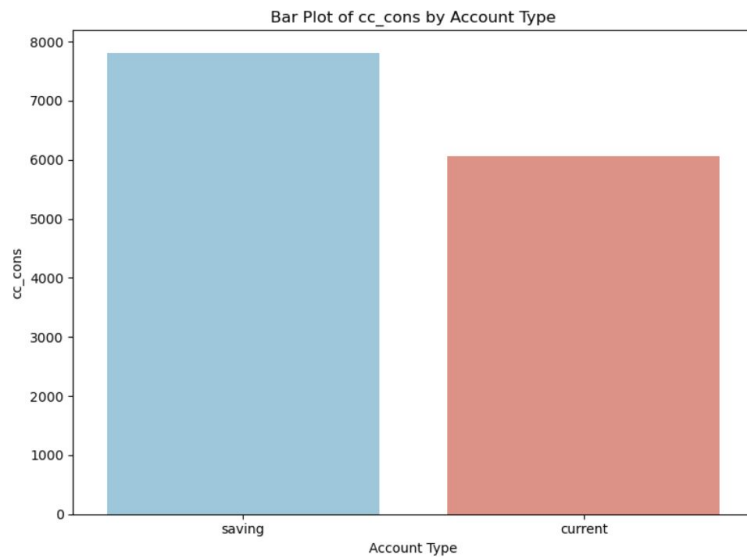
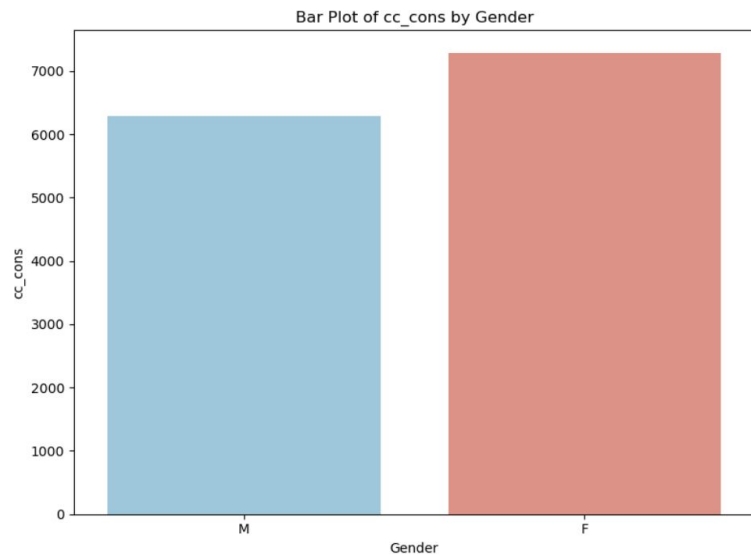
Visualizing and analyzing the dataset, detecting outliers, handling missing values, preprocessing

Initializing and training the dataset on various regression models and validating through the entire dataset.

The Features

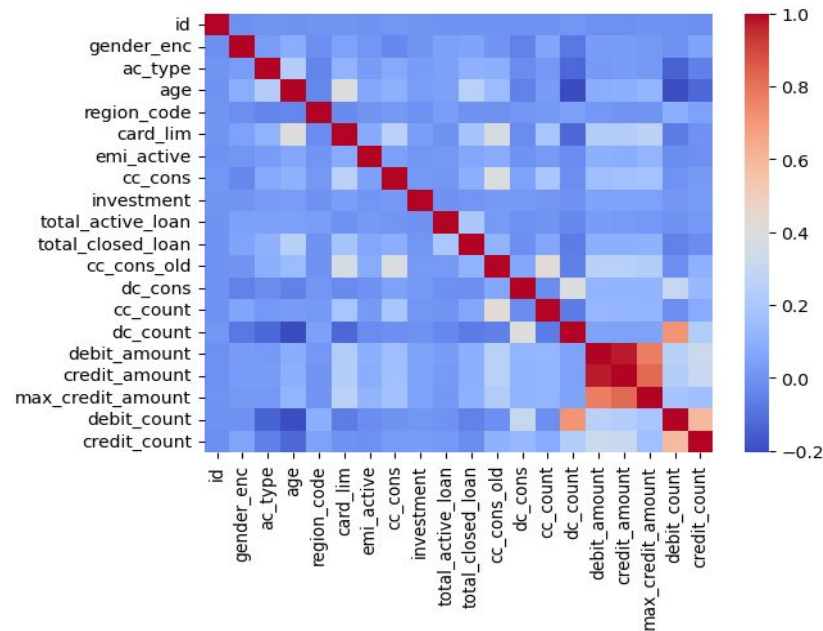
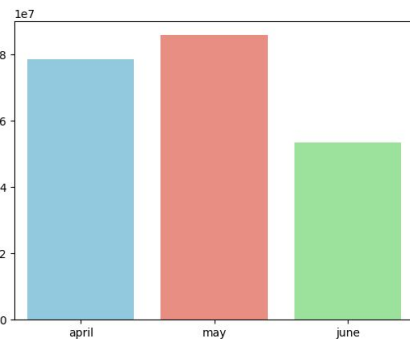
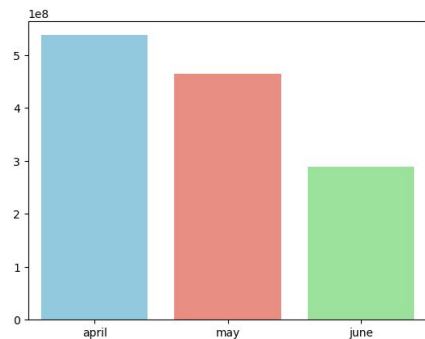


Visuals



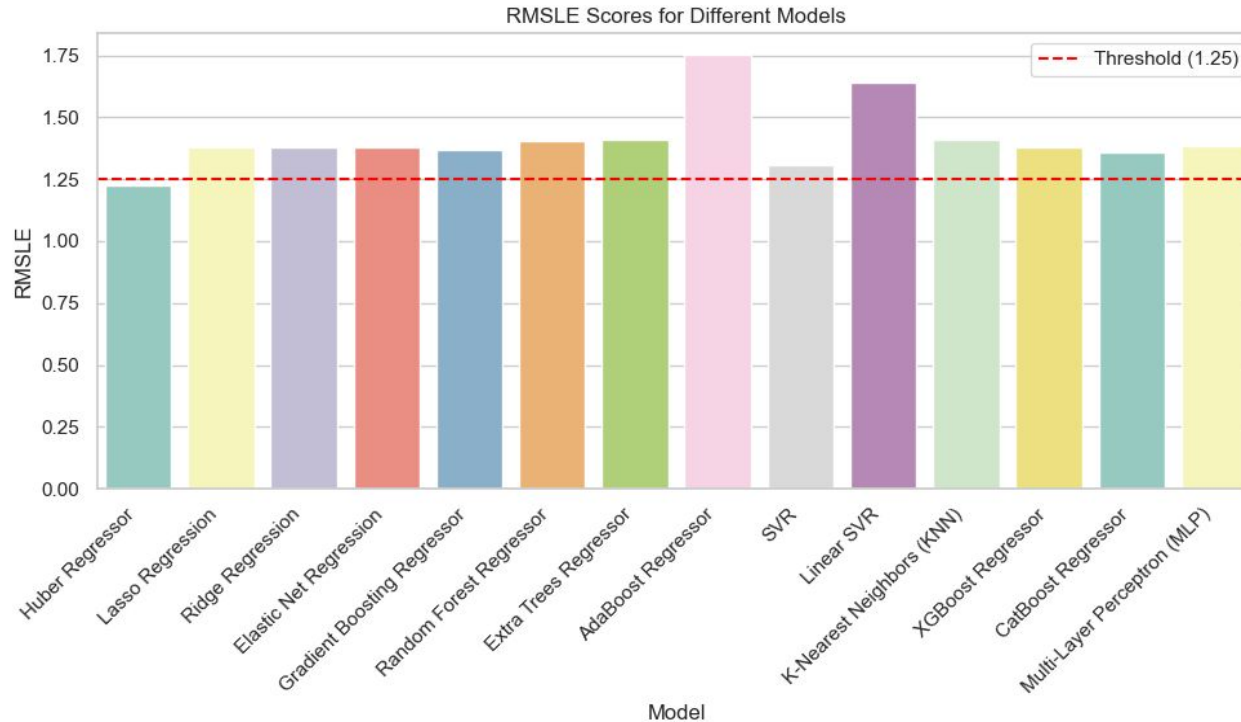
Target variable analysis on categorical features

Visuals



(L) Seasonal Expenditure Trends, Credit and Debit Card
(R) Correlation Heatmap of features after preprocessing

Visuals



Base model performances - 14 regressor models with different computation algorithms
Evaluation metric - root mean squared logarithmic error

Impact

Stacked Model Metrics:

Mean Squared Error:

106869096.3068

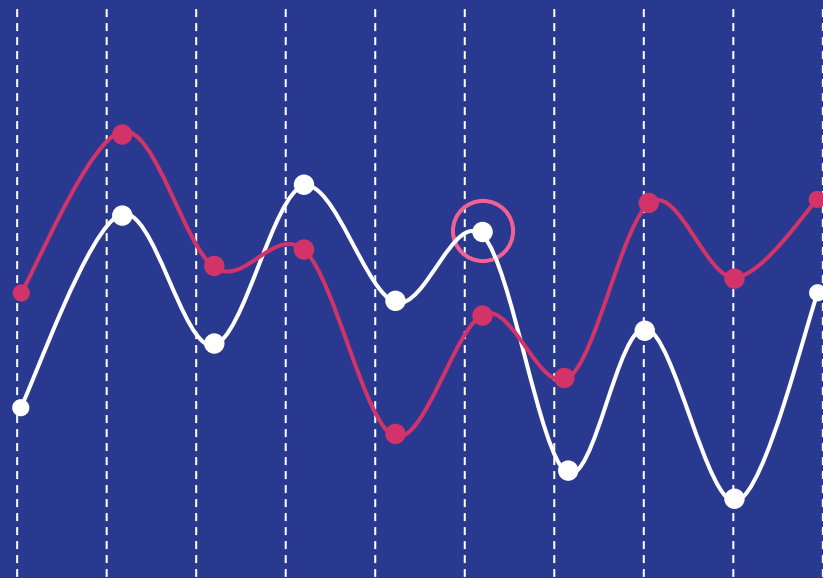
Mean Absolute Error: 4651.9976192

R2 Score: 0.160097

RMSLE: 1.11760446

RMSLE*100 on Validation Online:

116.1218436





Thank You